Text to SQL AI Agentic Workflow Implementations: A Systematic Literature Review

Eranda De Silva
Department of Industrial Management
University of Kelaniya
Sri Lanka
erandadesilva58@gmail.com

Dinesh Asanka
Department of Industrial Management
University of Kelaniya
Sri Lanka
dasanka@kln.ac.lk

2025 International Conference on Advances in Technology and Computing (ICATC) | 979-8-3315-9023-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICATC68823.2025.11407744

***Abstract*—This study comprehensively examines recent advancements and persistent challenges in the development of Text-to-SQL systems, specifically focusing on the integration of agentic workflows with a RAG or Graph-Retrieval Augmented Generation (Graph-RAG) approach. The core task of translating natural language queries into Structured Query Language (SQL) statements aims to democratize data access. However, existing methods, including standalone Large Language Models (LLMs), frequently exhibit limitations related to the ambiguity of natural language, the complexity of real-world database schemas, and the generation of factually incorrect or syntactically invalid SQL. This review synthesizes the literature to demonstrate how multi-agentic frameworks with RAG and Graph-RAG architectures address these issues. The decomposition of complex queries into sub-tasks including intent classification, schema linking, and error corrections posited to enhance the system's flexibility, generalization capabilities, and overall reliability. Concurrently, the RAG and Graph-RAG database context approach offers a promising method for grounding LLMs with high-fidelity, structured, and contextual information. By leveraging vector databases or knowledge graphs to represent database schemas and domain-specific knowledge, these RAG systems augment the LLM's prompt, thereby mitigating hallucinations and improving the accuracy of the generated SQL. TigerVector illustrates the potential of hybrid retrieval methods that combine semantic and relational search within a single graph database, enabling multi-hop reasoning crucial for handling complex enterprise and financial datasets. Despite these advancements, several challenges remain, notably the automation of creating effective, LLM-friendly database representations and achieving fully autonomous continual learning. Future research should focus on fortifying security protocols to facilitate the robust and scalable Text-to-SQL AI Agentic Graph-RAG workflows.**

Keywords—ai agents, graph-rag, large language models, rag, text-to-sql

I. Introduction

This systematic literature review is motivated by the growing demand for democratizing data access and streamlining data analysis across diverse domains, including finance, healthcare, and business intelligence [1]. The primary objective of Text-to-SQL systems is to translate natural language (NL) queries into accurate and executable Structured Query Language (SQL) statements, enabling direct interaction with databases for users without SQL proficiency [2]. This is a foundational task that aims to enhance business intelligence.

A. The Text-to-SQL Task and LLMs

The core goal of the Text-to-SQL task is to generate an accurate SQL query given the input natural language question and the database schema [3]. This capability holds immense promises for enabling non-technical users to effectively and intuitively retrieve information from databases. Further traditional Text-to-SQL methods often relied on parsing trees and human-crafted rules [4] or early sequence-to-sequence models [5].

However, the advent of Large Language Models (LLMs) has revolutionized this field, moving beyond these rule-based or simple translation approaches to more sophisticated, context-aware generation [4]. Despite this progress, LLMs face fundamental challenges in Text-to-SQL, particularly concerning scale, prompt engineering, and robustness [6]. These challenges include ambiguity in natural language, the complexity of real-world database schemas, and the generation of factually incorrect or syntactically invalid SQL.

B. Rationale and Persistent Limitations

Despite the critical importance of Text-to-SQL, significant gaps persist in the current state of the art. Traditional manual data access methods and early automated systems consistently demonstrate limitations, particularly with the inherent ambiguity of natural language and the complexity of real-world database schemas. These systems often struggle to generate syntactically correct and semantically meaningful SQL for complex operations, such as nested queries, joins, or aggregations. Further traditional Text-to-SQL methods often relied on parsing trees and human-crafted rules [4] or early sequence-to-sequence models [5].

Performance degradation is a notable concern, with reported accuracy declines of up to 30% when systems are applied to complex, real-world schemas compared to controlled academic benchmarks [7]. Furthermore, pervasive issues such as hallucinations (logically incorrect queries) and the high computational and API costs associated with Large Language Models (LLMs) have acted as substantial barriers to their practical adoption.

C. Transformative Methodological Paradigms

A transformative shift appears to be underway with the integration of two key methodological paradigms: AI agentic workflows and an RAG or Graph-Retrieval Augmented Generation (Graph-RAG) database context approach. Multi-agent Text-to-SQL framework [1] and the DB-GPT system [8] demonstrate how multiple specialized LLM are beneficial in constructing SQL queries from natural language.

AI Agentic workflows, such as those implemented in systems like MageSQL [1] and DB-GPT [8], orchestrate multiple specialized LLM agents in a coordinated pipeline. These multi-agent Text-to-SQL frameworks decompose complex queries into manageable sub-tasks. These sub-tasks typically include intent classification, schema linking, SQL

979-8-3315-9023-9/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.

construction, and error correction. Implementations, including the modular agent-based pipeline [1] illustrate that breaking down complex queries enhances flexibility, generalizability, and customizability [9]. Collectively, this decomposition not only improves performance but also makes LLM deployment more practical in real-world scenarios where schema diversity and query complexity are high. The synergy between AI agentic workflows and RAG or Graph-RAG represents a compelling solution for improving schema linking, handling complex queries, and achieving robust domain adaptation through dynamic context injection.

D. Objectives

The primary objective of this systematic literature review is to comprehensively evaluate the effectiveness and future trajectory of Text-to-SQL systems that integrate AI agentic workflows with a Retrieval Augmented Generation (RAG) database context approach. This review aims to address three key areas: the effectiveness of these combined methodologies, their adaptability in real-world applications, and the identification of future research gaps.

1) Effectiveness of AI Agentic Workflows and RAG Integration: This objective seeks to assess the performance and capabilities of contemporary AI agentic workflows that were observed in MageSQL [1], DB-GPT [8], OpenSearch-SQL [10] and their integration with a RAG database approach in Chat2Data [11], FinSQL [9], RBDQ [12] in enhancing Text-to-SQL translation. The evaluation will focus on how these advanced methods contribute to improved accuracy and robustness in generating SQL queries from natural language, particularly for complex queries involving multi-table joins, nested queries, and aggregations. This includes enhanced schema linking and understanding, especially with intricate or ambiguous database schemas. The review will also examine how these systems mitigate common LLM limitations, such as hallucinations, by providing relevant, structured, and contextual information, and how they improve domain generalization and adaptation to new databases.

2) Adaptability and Efficiency in Real-World Applications: This investigates the practical implications and suitability of AI agentic workflows and RAG or Graph-RAG [13] for deployment in real-world, high-stakes environments like finance, healthcare, and business intelligence [1]. The analysis will include an evaluation of the computational efficiency, processing speed, and latency of these systems to determine their viability for high throughput demands. The review will also investigate their capacity to support privacy-sensitive deployments and mitigate security vulnerabilities, such as prompt-to-SQL injections (P2SQL) [14]. Additionally, it will assess their ability to handle the challenges posed by long context lengths and imprecise question formulations common in real-world data.

3) Standardization of Methodologies and Future Research Gaps: This objective aims to identify critical gaps in current research and propose future directions for Text-to-SQL development, specifically for ai agentic and Graph-RAG-based [13] systems. This will involve a review of the variability in study designs and evaluation metrics in Spider [15] and BIRD [16] suggest more consistent and reliable methodologies. The review will explore ongoing challenges such as automating the creation of LLM-friendly database views, ensuring automated continual learning, and effectively integrating external knowledge [16]. Finally, it will investigate strategies for enhancing interpretability, debugging capabilities, and enabling human-in-the-loop validation to foster greater trust and flexibility in AI-generated SQL [4] [17].

II. METHODOLOGY

This study employed a comprehensive search strategy and a multi-stage manual selection process to identify and synthesize relevant studies on LLM-based Text-to-SQL systems.

A. Eligibility Criteria

1) Inclusion Criteria: The review included studies specifically investigating the application of Large Language Models (LLMs) in Text-to-SQL systems. A particular focus was placed on the integration of AI agentic workflows seen in MageSQL [1], DB-GPT [8], OpenSearch-SQL [10], A Modular Agent-Based Approach and Retrieval-Augmented Generation (RAG) approaches, especially those leveraging Graph-RAG techniques for database context as in graph database approaches in KET-RAG [18] and TigerVector [19]. Research covering related challenges, such as schema linking, query complexity in multi-table joins, nested queries, hallucinations, and security vulnerabilities as identified in, P2SQL injections [14], were also included. The Publication Date was restricted to studies from January 2023 to the present, and only peer-reviewed journal articles and conference papers published in English were considered. Included studies had to report key outcomes such as execution accuracy (EX) [20] and exact match (EM) [21].

2) Exclusion Criteria: Studies that did not primarily focus on LLM-based Text-to-SQL systems, or those utilizing traditional non-LLM methods, were excluded. Publications outside the specified date range (January 2023 to present) were excluded. Non-peer-reviewed articles, preprints not sourced from arXiv, dissertations, meta-analyses, editorials, and commentaries were excluded to maintain focus on high-quality, original research.

B. Grouping for Synthesis

The studies were categorized into six thematic groups to facilitate a comprehensive and structured synthesis of the research landscape. Fundamental LLM-based Text-to-SQL Paradigms group focuses on core components such as Semantic Parsing and Schema Linking. Advanced Architectural Frameworks and Agentic Workflows included research on modular or multi-agent systems designed to handle intricate queries, such as MageSQL [1], DB-GPT [8], OpenSearch-SQL [10], and COGSQL [22]. Contextual Augmentation with Retrieval-Augmented Generation (RAG) and Graph Databases are studies leveraging external structured context, including Graph-RAG related studies observed in KET-RAG [18] and TigerVector [19], aimed at mitigating LLM hallucinations. Performance Evaluation, Reliability, and Security addresses metrics, benchmarks like Spider, BIRD, error detection, and defenses against Prompt-to-SQL (P2SQL) injection attacks [14] are observed in the studies. Domain Adaptation, Generalization, and Efficiency Optimization studies cover techniques like fine-tuning small

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.

LLMs and optimizing performance for efficiency and scalability. User Interaction, Interpretability, and Human-in-the-Loop Systems highlighted efforts for transparency, including visual analysis tools as discussed in MAVIDSQL [23] and human-in-the-loop validation strategies [17].

C. Information Sources and Search Strategy

Databases utilized included IEEE Xplore, ACM Digital Library, Scopus, and Google Scholar**.** Specialized platforms consulted included ResearchGate, arXiv (for preprints), Hugging Face, GitHub (for frameworks like DB-GPT and MageSQL), and dedicated Benchmark Websites/Leaderboards like Spider and BIRD.

The search was conducted on July 27, 2025. Search terms combined "Text-to-SQL" (or variants like "nl-to-sql"), "LLM" (or "Large language Models"), and "ai agent" (or "multi agent" or "ai agent workflow"). For Google Scholar, "RAG" or "GRAPHRAG" terms were also included. Filters restricted the search to the 2023 to present date range, English language, and publication types limited to peer-reviewed journal articles, conference proceedings, and arXiv preprints.

D. Selection Process

The search initially yielded approximately 550 records. A first-level manual screening of titles and abstracts retained approximately 150 records, including those with no citations to capture emerging research. Subsequent detailed abstract review resulted in a pool of approximately 84 records. Full-text reviews confirmed direct relevance to LLM-based Text-to-SQL systems incorporating AI agents or RAG frameworks, culminating in a final set of 43 articles included in the review. The selection process was conducted manually without automation tools to ensure meticulousness.

E. Data Collection and Items

Data were extracted using a standardized form, focusing on key items necessary for holistic evaluation. In the studies LLMs utilized included models such as GPT-4o [13], Llama 3 [24], Gemma-2B [25], Claude 3 [26], and Phi-3 [27]. Studies which highlighted Methodologies and Frameworks which are focused on Multi-Agent Systems are seen in MageSQL [1] and DB-GPT [8], Prompt Engineering Strategies in PTD-SQL [2], Schema Linking methods in PS-SQL [28], RESDSQL [29], and Retrieval-Augmented Generation (RAG) techniques were seen in studies as Chat2Data [11], KET-RAG [18]. As per the studies evaluation and performance documented datasets and benchmarks such as Spider, BIRD, WikiSQL and metrics such as Execution Accuracy (EX) and Exact Match (EM).

Included studies by publication year for this review shows a peak in 2025 reflecting rapid growth in comparison to 2023 when it comes to studies on Text-to-SQL.

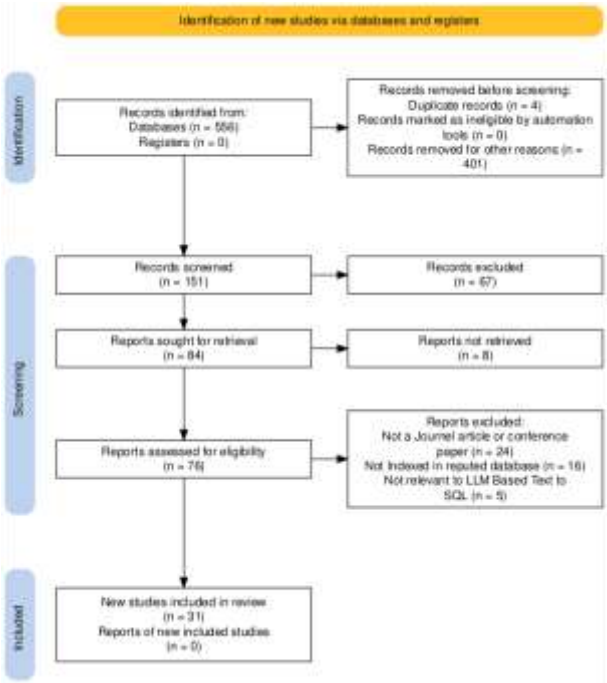


Fig. 1. Prisma flow diagram

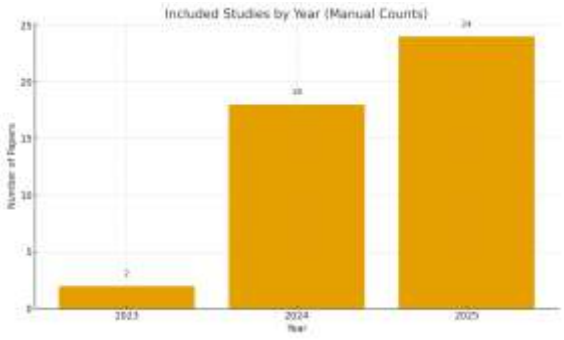


Fig. 2. Included studies by publication year (2023 -2025)

III. Discussion

A. Synthesis of Findings

This systematic review reveals a clear trajectory in the Text-to-SQL field toward enhancing Large Language Models (LLMs) with structured context and modular reasoning. The findings demonstrate that AI agentic workflows with RAG and Graph-Retrieval Augmented Generation (Graph-RAG) approaches are complementary solutions to the longstanding challenges of natural language-to-SQL translation.

Multi-agent frameworks significantly improve the handling of query complexity and ambiguity by delegating tasks to specialized agents. Concurrently, retrieval-augmented methods enhance factual accuracy by grounding the LLM in the actual database schema and content [12]. The resulting combined effect is a notable increase in robustness, particularly for complex queries involving multiple tables, intricate conditions, or uncommon domain terminology.

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.

1) Fundamental LLM-based Text-to-SQL Paradigms: The core evolution involves transforming the input prompt to guide LLM reasoning effectively, particularly concerning few-shot examples and structured output. Multiple studies focus on optimizing prompt engineering for LLMs. DAIL-SQL systematically evaluated various components, including question representation, example selection, and organization across multiple LLMs, to refresh performance benchmarks [6]. SC-Prompt utilized a divide-and-conquer framework for few-shot learning, explicitly separating the generation process into a structure stage (generating SQL commands with placeholders) and a content stage (populating placeholders with identifiers and values) [30].

To handle the complexity of intertwining schema items and SQL syntax, frameworks advocate for decoupling the process. RESDSQL proposed decoupling schema linking and skeleton parsing by using a ranking-enhanced encoder to inject only the most relevant schema items, and a skeleton-aware decoder that first generates the SQL skeleton to implicitly constrain subsequent SQL generation [29]. PS-SQL further enhanced schema linking accuracy by incorporating phrase structure information (Noun Phrases) from the natural language question to improve alignment between question tokens and schema items [28].

2) Advanced Architectural Frameworks and Agentic Workflows: The current literature moves beyond single-step translation to integrate sophisticated workflows that mimic human problem-solving and verification. Foundational decomposition approaches like DIN-SQL break the Text-to-SQL task into phased subtasks (schema linking, question classification/decomposition, SQL generation, and self-correction) [4], [5]. PTD-SQL extends this by applying Partitioning and Targeted Drilling, grouping questions based on logical thinking paths (Query Grouping Principles, QGP), allowing the LLM to specialize its reasoning for complex query groups [2]. Similarly, QC-SQL enhances generation accuracy by dynamically applying different strategies based on a Query Classification module [5].

The complexity of multi-agent systems introduces instability [10]. OpenSearch-SQL addresses this by proposing a standard four-stage framework (Preprocessing, Extraction, Generation, and Refinement) enhanced with a **Consistency** Alignment mechanism. This alignment mechanism aims to stabilize the output of specialized agents and reduce the accumulation of hallucinations throughout the workflow. Furthermore, OpenSearch-SQL utilizes an intermediate language, SQL-Like, within its structured Chain-of-Thought logic to encourage LLMs to focus on the logical backbone of the query before specific syntax.

Agent frameworks are increasingly equipped with specialized tools for verification. ToolSQL introduced a tool-assisted agent framework, converting SQL generation into a Python generation task to enable the LLM agent to leverage external tools like a Database Retriever and an Error Detector for SQL verification and refinement [21]. This system specifically targets database mismatches (errors that do not trigger execution exceptions, such as condition mismatches) prevalent in real-world scenarios.

Enterprise systems like SQLord [31] and RBDQ [12], which blend workflow decomposition with knowledge augmentation, achieved production-level performance that far exceeds what earlier single-shot neural parsers could deliver. The importance of domain context and knowledge is a unifying theme, as traditional models often fail when a query implies unstated knowledge in scenarios such as default filtering or specific column references. Retrieval-based systems, such as Chat2Data [11], explicitly address this by fetching relevant schema elements or previous queries, ensuring the LLM does not rely purely on learned prior knowledge, which reduces hallucinations and syntax errors.

The synergy between agentic decomposition (reasoning) and context injection via retrieval or graphs (knowledge grounding) is powerful. However, this specialization introduces a trade-off when systems tailored to a domain as explored in FinSQL [9] or curated knowledge base report the highest accuracies, but they may exhibit brittleness when encountering inputs outside their design assumptions.

B. Comparative Overview of Approaches

TABLE I. COMPARISON OF APPROACHES

Study	**Core Method**	**Key Limitations & Critical Gaps**
MageSQL [1], DB-GPT [8]	Multi-Agent Architectures which decompose tasks (e.g., schema linking, error correction) to specialized agents.	Significant overhead due to multiple sequential LLM calls. Risk of error propagation between agent stages. Higher token consumption per query compared to single pass.
Chat2Data [11], RBDQ [12].	RAG Retrieves relevant schema context/rows to ground the LLM and reduce hallucinations.	Requires complex external knowledge stores/indices. Performance is strictly bound by retrieval accuracy (retrieval errors lead to SQL errors).
KET-RAG [18]	Graph-RAG Uses structured knowledge graphs for multi-hop reasoning and context.	High computational cost for maintaining and traversing large-scale graphs. Labor-intensive initial knowledge base construction.
FinSQL [9], SQLord [31]	Fine-Tuned Domain Models. Customizes smaller models (e.g., Llama, Qwen-7B) on domain-specific datasets.	Sharp performance drops on unseen schemas or new domains. Requires large, high-quality annotated training datasets which are scarce.
GPT-4o [13]	General / Closed-Book Prompting. Relies on few-shot prompting without external structural grounding.	Lacks robustness for complex nested queries. Prone to inventing schema elements when context is exceeded.

The reviewed studies present distinct strategies for Text-to-SQL generation, each with unique advantages in each approach given the capability of the design and implementations.

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.

C. Challenges and Future Directions

Despite significant progress, several ongoing challenges persist, suggesting key areas for future research and development in Text-to-SQL systems and application development.

1) Schema Linking and Knowledge Base Construction: The current process of creating LLM-friendly schema representations is often labor-intensive and relies on static descriptions. Robust schema-grounding techniques are required as seen in approaches for GraphRAG in KET-RAG [18], as current models, including strong LLMs like Falcon-7B, still make serious mistakes regarding database schema [15]. Advanced indexing, like the multi-granular indexing seen in KET-RAG, needs to be scaled up to handle larger schemas effectively.

2) Continual Learning and Adaptability: Most reviewed systems are static after design, meaning they do not autonomously improve over time as schemas or user query styles evolve. Systems should incorporate fully autonomous continual learning, learning from each user interaction, potentially by using self-reflection mechanisms as identified in studies like RBDQ [12].

3) Integrating External and Unstructured Knowledge: A major gap is the inability of current systems to handle questions requiring external knowledge beyond the database. Graph-RAG is naturally suited for this integration, as it can link SQL schema nodes with general domain knowledge nodes.

4) Efficiency, Scalability, and Cost: Running multi-agent pipelines or accessing large knowledge bases incurs significant latency and computational cost, acting as a barrier to enterprise adoption. Prioritizing efficiency through the development of lighter-weight LLMs specialized for sub-tasks, as well as systematic coaching and result reuse as observed in the vector DB cache in Chat2Data [11] study.

5) Evaluation Standardization and Benchmarking: Current evaluation suffers from a lack of standardized, comprehensive benchmarks. Evaluations must move beyond simple metrics like exact string match toward user-centric evaluation and incorporating criteria from complex benchmarks like BIRD. Consistent frameworks, such as SQLord using its GPT-Judge [31], are needed to holistically evaluate SQL correctness, semantic consistency, and optimality.

6) Reliability, Interpretability, and Security: For broad adoption, systems must be trustworthy and secure. Reliability requires embedding verification steps and self-reflection mechanisms. Interpretability can be enhanced by providing natural language explanations alongside the generated SQL, increasing user trust. On the security front, safeguards against Prompt-to-SQL (P2SQL) injections and sensitive data leakage are critical [14].

IV. Conclusion

In this systematic literature review, we examined the state-of-the-art advancements in Text-to-SQL systems, focusing particularly on the integration of AI agentic workflows with RAG or Graph-Retrieval-Augmented Generation (Graph-RAG) database context approaches. The findings underscore a transformative shift in the field, moving beyond single-step query parsing to harness modular reasoning and structured knowledge retrieval.

The synthesis of evidence from 31 recent studies confirms that these combined approaches substantially improve the accuracy, robustness, and practicality of natural language to SQL translation. Multi-agent frameworks, such as DB-GPT and MageSQL, are effective in orchestrating complex query generation by decomposing tasks (intent, schema linking, error correction), leading to more reliable outputs. Concurrently, the adoption of RAG and Graph-RAG as observed in Chat2Data, RBDQ, and FinSQL has markedly reduced factual errors (hallucinations) and improved execution accuracy by grounding LLMs in the relevant schema and domain content. This confluence of agentic reasoning and retrieval grounding has resulted in record-breaking performance, with enterprise-focused solutions attaining over 90% correctness in online production use. Furthermore, systems like SQLord, RBDQ, have demonstrated practical implementation, signalling a maturing technology ready for wider adoption. Despite this progress, the review highlights persistent open challenges before Text-to-SQL agents can be considered a solved problem. These include achieving robust generalization to entirely new domains, efficient scaling to massive knowledge bases, and the seamless integration of external, unstructured knowledge sources. The complexity inherent in multi-agent systems also introduces challenges related to development overhead and potential error propagation between agents. To fully realize the vision of democratized data access, future research must build on these identified trends. Future work must focus on automating knowledge base construction for instance, using LLMs to generate a knowledge graph of the schema. Research is needed into safe model updating methods, such as episodic memory or meta-learning, to ensure evolution without catastrophic failure of past capabilities of SQL construction. Future Text-to-SQL agents need to be linked with web search or documentation retrieval agents to handle combined queries. Cost-awareness, including dynamically allocating resources based on query complexity, should be built into agent planning. Tackling security aspects in these systems involves techniques like prompt filtering, role-based access control for agents, and monitoring for anomalous instructions by a supervisor agent should also be focused on future systems. Efforts should focus on unifying the strengths of current approaches into a coherent system that can dynamically decide when to invoke an agent, retrieve knowledge, or ask for clarification. Critical areas for continued investigation include automating knowledge integration, enabling lifelong learning, improving efficiency, and strengthening trustworthiness, interpretability, and security in enterprise deployments. Ultimately, these steps will facilitate the development of the next generation of intelligent, context-aware SQL generation agents.

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] C. Shen, J. Wang, S. Rahman, and E. Kandogan, "Demonstration of a Multi-agent Framework for Text to SQL Applications with Large Language Models," in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, Boise ID USA: ACM, Oct. 2024, pp. 5280–5283. doi: 10.1145/3627673.3679216.

[2] R. Luo, L. Wang, B. Lin, Z. Lin, and Y. Yang, "PTD-SQL: Partitioning and Targeted Drilling with LLMs in Text-to-SQL," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Miami, Florida, USA: Association for Computational Linguistics, 2024, pp. 3767–3799. doi: 10.18653/v1/2024.emnlp-main.221.

[3] S. Chafik, S. Ezzini, and I. Berrada, "Towards Automating Domain-Specific Data Generation for Text-to-SQL: A Comprehensive Approach," *ACM Trans. Softw. Eng. Methodol.*, p. 3746226, June 2025, doi: 10.1145/3746226.

[4] K. Chen, Y. Chen, N. Koudas, and X. Yu, "Reliable Text-to-SQL with Adaptive Abstention," *Proc. ACM Manag. Data*, vol. 3, no. 1, pp. 1–30, Feb. 2025, doi: 10.1145/3709719.

[5] R. Gao, J. Guo, H. Zhao, J. Diao, Q. Lu, and Q. Peng, "Enhanced Text-to-SQL Generation via Query Classification," in *Frontiers in Artificial Intelligence and Applications*, A. J. Tallón-Ballesteros, Ed., IOS Press, 2025. doi: 10.3233/FAIA250441.

[6] D. Gao *et al.*, "Text-to-SQL Empowered by Large Language Models: A Benchmark Evaluation," *Proc. VLDB Endow.*, vol. 17, no. 5, pp. 1132–1145, Jan. 2024, doi: 10.14778/3641204.3641221.

[7] M. Ganti, L. Orr, and S. Wu, "Evaluating Text-to-SQL Model Failures on Real-World Data," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, Utrecht, Netherlands: IEEE, May 2024, pp. 1–1. doi: 10.1109/ICDE60146.2024.00456.

[8] S. Xue *et al.*, "Demonstration of DB-GPT: Next Generation Data Interaction System Empowered by Large Language Models," *Proc. VLDB Endow.*, vol. 17, no. 12, pp. 4365–4368, Aug. 2024, doi: 10.14778/3685800.3685876.

[9] C. Zhang *et al.*, "FinSQL: Model-Agnostic LLMs-based Text-to-SQL Framework for Financial Analysis," in *Companion of the 2024 International Conference on Management of Data*, Santiago AA Chile: ACM, June 2024, pp. 93–105. doi: 10.1145/3626246.3653375.

[10] X. Xie, G. Xu, L. Zhao, and R. Guo, "OpenSearch-SQL: Enhancing Text-to-SQL with Dynamic Few-shot and Consistency Alignment," *Proc. ACM Manag. Data*, vol. 3, no. 3, pp. 1–24, June 2025, doi: 10.1145/3725331.

[11] X. Zhao, X. Zhou, and G. Li, "Chat2Data: An Interactive Data Analysis System with RAG, Vector Databases and LLMs," *Proc. VLDB Endow.*, vol. 17, no. 12, pp. 4481–4484, Aug. 2024, doi: 10.14778/3685800.3685905.

[12] F. Bi *et al.*, "RBDQ: A Reliable LLM-based Text-to-SQL System for Business Data Queries," in *Companion Proceedings of the ACM on Web Conference 2025*, Sydney NSW Australia: ACM, May 2025, pp. 95–103. doi: 10.1145/3701716.3715257.

[13] A. Krishna, A. Shinde, S. Ardham, and C. Malhotra, "A Collaborative Multi-Agent LLM Approach for Knowledge Graph Curation and Query from Multi-Modal Data Sources," Feb. 06, 2025, *Preprints*. doi: 10.36227/techrxiv.173886317.71891930/v1.

[14] R. Pedro, M. E. Coimbra, D. Castro, P. Carreira, and N. Santos, "Prompt-to-SQL Injections in LLM-Integrated Web Applications: Risks and Defenses," in *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, Ottawa, ON, Canada: IEEE, Apr. 2025, pp. 1768–1780. doi: 10.1109/ICSE55347.2025.00007.

[15] V. Câmara, R. Mendonca-Neto, A. Silva, and L. Cordovil, "A Large Language Model approach to SQL-to-Text Generation," in *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA: IEEE, Jan. 2024, pp. 1–4. doi: 10.1109/ICCE59016.2024.10444148.

[16] A. Singh, A. Shetty, A. Ehtesham, S. Kumar, and T. T. Khoei, "A Survey of Large Language Model-Based Generative AI for Text-to-SQL: Benchmarks, Applications, Use Cases, and Challenges," in *2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA: IEEE, Jan. 2025, pp. 00015–00021. doi: 10.1109/CCWC62904.2025.10903689.

[17] Y. Tian *et al.*, "Text-to-SQL Domain Adaptation via Human-LLM Collaborative Data Annotation," in *Proceedings of the 30th International Conference on Intelligent User Interfaces*, Cagliari Italy: ACM, Mar. 2025, pp. 1398–1425. doi: 10.1145/3708359.3712083.

[18] Y. Huang, S. Zhang, and X. Xiao, "KET-RAG: A Cost-Efficient Multi-Granular Indexing Framework for Graph-RAG," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, Toronto ON Canada: ACM, Aug. 2025, pp. 1003–1012. doi: 10.1145/3711896.3737012.

[19] S. Liu *et al.*, "TigerVector: Supporting Vector Search in Graph Databases for Advanced RAGs," in *Companion of the 2025 International Conference on Management of Data*, Berlin Germany: ACM, June 2025, pp. 553–565. doi: 10.1145/3722212.3724456.

[20] L. Yan, Q. Wan, C. Liu, S. Duan, P. Han, and Y. Xu, "SPS-SQL: Enhancing Text-to-SQL generation on small-scale LLMs with pre-synthesized queries," *Pattern Recognit. Lett.*, vol. 196, pp. 45–51, Oct. 2025, doi: 10.1016/j.patrec.2025.04.016.

[21] Z. Wang, R. Zhang, Z. Nie, and J. Kim, "ToolSQL: A Tool-Assisted Agent for SQL Verification and Refinement," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, Toronto ON Canada: ACM, Aug. 2025, pp. 3102–3110. doi: 10.1145/3711896.3737159.

[22] H. Yuan, X. Tang, K. Chen, L. Shou, G. Chen, and H. Li, "CogSQL: A Cognitive Framework for Enhancing Large Language Models in Text-to-SQL Translation".

[23] J. Tang *et al.*, "MAVIDSQL: A Model-Agnostic Visualization for Interpretation and Diagnosis of Text-to-SQL Tasks," *IEEE Trans. Cogn. Dev. Syst.*, vol. 16, no. 5, pp. 1887–1903, Oct. 2024, doi: 10.1109/TCDS.2024.3391278.

[24] M. Lashuel, G. Kurdistan, A. Green, J. S. Erickson, O. Seneviratne, and K. P. Bennett, "LLM - Based Code Generation for Querying Temporal Tabular Financial Data," in *2024 IEEE Symposium on Computational Intelligence for Financial Engineering and Economics (CIFEr)*, Hoboken, NJ, USA: IEEE, Oct. 2024, pp. 1–8. doi: 10.1109/CIFEr62890.2024.10772804.

[25] H. Neogi, J. Ray, and S. Banerjee, "Fine-Tuning Gemma-2B for Text-to-SQL Generation," in *2024 9th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India: IEEE, Dec. 2024, pp. 265–269. doi: 10.1109/ICCES63552.2024.10860111.

[26] M. Urban, J. Ding, D. Kernert, K. Vaidya, and T. Kraska, "Utilizing Past User Feedback for More Accurate Text-to-SQL," in *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, Intercontinental Berlin Berlin Germany: ACM, June 2025, pp. 1–7. doi: 10.1145/3736733.3736739.

[27] E. Öztürk, "Improving Text-to-Sql Conversion for Low-Resource Languages Using Large Language Models," *Bitlis Eren Üniversitesi Fen Bilim. Derg.*, vol. 14, no. 1, pp. 163–178, Mar. 2025, doi: 10.17798/bitlisfen.1561298.

[28] Z. Lan and S. Li, "PS-SQL: Phrase-based Schema-Linking with Pre-trained Language Models for Text-to-SQL Parsing," in *2024 6th International Conference on Natural Language Processing (ICNLP)*, Xi'an, China: IEEE, Mar. 2024, pp. 31–35. doi: 10.1109/ICNLP60986.2024.10692964.

[29] H. Li, J. Zhang, C. Li, and H. Chen, "RESDSQL: Decoupling Schema Linking and Skeleton Parsing for Text-to-SQL," *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 11, pp. 13067–13075, June 2023, doi: 10.1609/aaai.v37i11.26535.

[30] Z. Gu *et al.*, "Few-shot Text-to-SQL Translation using Structure and Content Prompt Learning," *Proc. ACM Manag. Data*, vol. 1, no. 2, pp. 1–28, June 2023, doi: 10.1145/3589292.

[31] S. Cheng, Q. Cheng, L. Jin, L. Yi, and G. Zhang, "SQLord: A Robust Enterprise Text-to-SQL Solution via Reverse Data Generation and Workflow Decomposition," in *Companion Proceedings of the ACM on Web Conference 2025*, Sydney NSW Australia: ACM, May 2025, pp. 919–923. doi: 10.1145/3701716.3715541.

Authorized licensed use limited to: ULAKBIM UASL - SABANCI UNIVERSITY. Downloaded on October 12,2026 at 15:23:21 UTC from IEEE Xplore. Restrictions apply.